

(1001CJA621620025)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

UNIT TEST

02-08-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A,D	A,B,C	A,B,C	A,B,C,D	B,C	B	D	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	8.28	4.90 to 5.10	16.00	16.00	6.00	1.68 to 1.69	60.00	6.40		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	A,D	A,D	A,B,D	A,B,C,D	A,C,D	B	A	B	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	240.00	-450.00	-3.15	0.30	1.50	1.50	2.00	3.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	C	A,D	A,B,C,D	C	A,B,C	B	C	C	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	1.00	0.50	-16.00	0.66 or 0.67	1.61 or 1.62	2.00	9.00	2007.00		

(1001CJA621620026)

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**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Main)

UNIT TEST

02-08-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	C	C	B	C	B	D	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	B	D	C	B	A	C	C	A
SECTION-II	Q.	1	2	3	4	5					
	A.	2.50	0.64	0.75	6.00	138.39					

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	A	C	C	C	B	B	C	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	C	D	C	C	D	A	B	C
SECTION-II	Q.	1	2	3	4	5					
	A.	15.00	4.00	5.00	0.00	9.00					

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	B	D	A	A	C	B	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	B	B	A	B	B	C	A	A
SECTION-II	Q.	1	2	3	4	5					
	A.	0.50	3.50	0.50	2.00	0.80					

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I****1. Ans. (A,C)****Sol.** Magnetic field inside the cavity remains constant

$$\vec{B}_{\text{cavity}} = \frac{\mu_0 \vec{J} \times \vec{\ell}}{2}$$

$$J = \frac{I}{A} = \frac{I}{\pi R^2 - \pi \left(\frac{R}{3}\right)^2} = \frac{I}{\frac{8\pi R^2}{9}}$$

Let say,

 $\vec{B}_1$ = Magnetic field due to large cylinder. $\vec{B}_2$ = Magnetic field due to small cylinder.

$$\vec{B}_B = \vec{B}_1 + \vec{B}_2$$

$$= \frac{\mu_0 J \times 2R/3}{2} + 0 = \frac{3\mu_0 I}{8\pi R}$$

Magnetic field inside cavity is constant so

$$\vec{B}_A = \vec{B}_B$$

2. Ans. (A,D)**Sol.** From circuit analysis

$$\text{Key open : } C_{\text{eq}} = 5 \mu\text{F} \Rightarrow Q = 50 \mu\text{C}$$

$$\text{Key closed : } C_{\text{eq}} = 10 \mu\text{F} \Rightarrow Q = 100 \mu\text{C}$$

$$Q_{\text{flowed}} = 50 \mu\text{C}$$

$$U_i = \frac{1}{2} (5)(10)^2 = 250$$

$$U_f = \frac{1}{2} (10)(10)^2 = 500$$

$$\Delta U = 250 \mu\text{J}$$

3. Ans. (A,B,C)**Sol.** For mean position $\rho = \rho_1 \Rightarrow kh = \rho_1$

$$h = \frac{\rho_1}{k}$$

initially particle was at extreme

$$\text{hence } A = \frac{\rho_1}{k}$$

Max kinetic energy

$$= \frac{1}{2} kA^2 \text{ where } k \text{ is force constant}$$

4. Ans. (A,B,C)

$$\text{Sol. } dF = i(d\vec{\ell} \times \vec{B})$$

at all point of circumference

here $d\vec{\ell} \parallel \vec{B}$, hence $F = 0$ **5. Ans. (A, B, C, D)****Sol.** Steady-state capacitor behave as open circuit

So 1 A current will flow in all resistances

From given circuit

$$V_{2\mu\text{F}} = V_{3\Omega} = 3\text{V}$$

$$\therefore q_{2\mu\text{F}} = 2 \times 3 = 6 \mu\text{C}$$

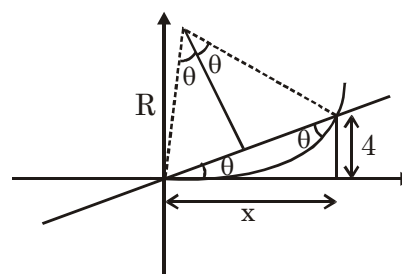
$$\text{Also } V_{4\mu\text{F}} = V_{4\Omega} = 4\text{V}$$

$$\therefore U = \frac{1}{2} \times 4\mu\text{F} \times (4\text{V})^2 = 32\mu\text{J}$$

6. Ans. (B,C)**Sol.** From given diagram, particle will rotated through an angle "2θ" as shown

$$\therefore t = 2\theta \frac{m}{qB_0}$$

$$x = R \sin 2\theta, y = R(1 - \cos 2\theta)$$



It will inter at A come out from B.

From geometry $AB = 2R\sin\theta$

where R is radius of curvature

$$R = \frac{mV}{qB}$$

$$x = AB\cos\theta, y = AB\sin\theta, t = 0$$

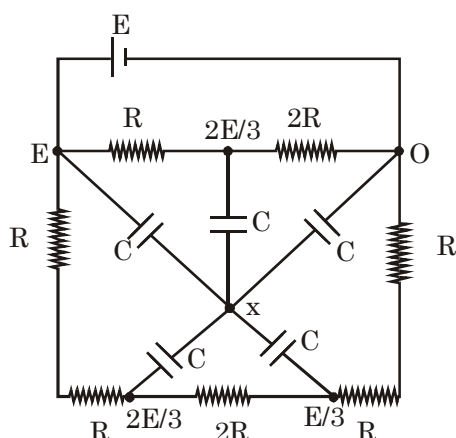
$$= 2R\sin\theta\cos\theta = 2R\sin\theta\sin\theta, t = 0$$

$$(x = \frac{mV}{qB}\sin 2\theta, y = R(1 - \cos 2\theta) t = 0$$

7. Ans. (B)

8. Ans. (D)

Sol. (7 and 8)



Charge of an isolated system is conserved,
From circuit analysis, potential of different points is as shown let potential of junctions be 'x'

$$(x - E) + 2(x - \frac{2E}{3}) + x - \frac{E}{3} + x = 0$$

$$\therefore x = \frac{8E}{15}$$

$\therefore$ Charge on capacitor

$$1 = \left(E - \frac{8E}{15}\right)C = \frac{7CE}{15}$$

$$\text{and } v_A - v_B = \frac{E}{6}$$

9. Ans. (B)

Sol. Initially water level will rise upto h_0 in tube then both will fill up so slope or rate of rise of h_1 decreases and finally becomes h_1

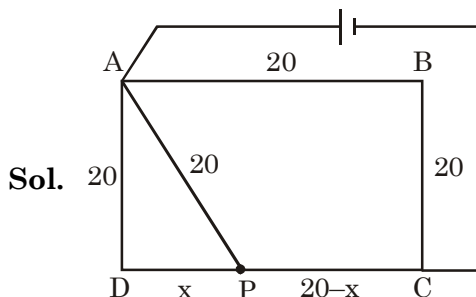
10. Ans. (C)

Sol. Initially water will only fill the capillary. Then slowly the water will rise up to h_1 and due to pressure the meniscus will invert and water will rise to h_1 & h_0 .

then it will start overflow & becomes constant

SECTION-II

1. Ans. 8.28



Sol.

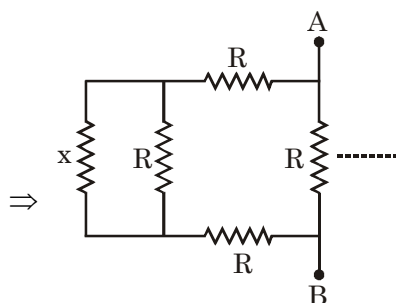
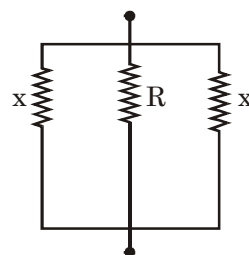
Since B and P are at same potential

$$R_{AP} = R_{PC}$$

$$\frac{20(20+x)}{20+20+x} = 20-x$$

2. Ans. 4.90 to 5.10

Sol. From infinite series



$$\frac{xR}{R+x} + 2R = x$$

$$xR + 2R^2 + 2Rx = xR + x^2$$

$$x^2 - 2Rx - 2R^2 = 0$$

$$x = \left(\frac{2 + \sqrt{4 + 8}}{2}\right)R$$

$$x = (\sqrt{3} + 1)R$$

$$\frac{1}{R_{eq}} = \frac{1}{x} + \frac{1}{R} + \frac{1}{x}$$

$$R_{eq} = 100 \Omega$$

3. **Ans. 16.00**

Sol. From torricelli theorem

$$v = \sqrt{2g(20)}$$

$$R = \sqrt{2g(20)} \sqrt{\frac{2 \times 80}{g}}$$

$$= 2 \times 40 = 80 \text{ m}$$

$$R' = 3 \times 80 = 240 \text{ m} = v' \sqrt{\frac{2 \times 80}{10}}$$

$$240 = v'(4)$$

$$v' = 60 \text{ m/s}$$

$$P_0 + \frac{F}{A} = P_0 + \frac{1}{2} \rho v'^2$$

4. **Ans. 16.00**

Sol. Using symmetry

$$R_{eq} = \frac{R}{8} + \frac{R}{16} = \frac{3R}{16}$$

5. **Ans. 6.00**

Sol. From property of ellipse

$$a - ae = r_0$$

$$a + ae = r_1$$

$$a = \frac{r_0 + r_1}{2}$$

$$e = \frac{r_1 - r_0}{r_1 + r_0}$$

$$\text{Radius of curvature} = \frac{b^2}{a} = a^2 \times \frac{1 - e^2}{a} =$$

$$\frac{2r_0r_1}{r_0 + r_1}$$

$$\alpha = 6$$

6. **Ans. 1.68 to 1.69**

$$\text{Sol. } r = \left(\frac{\ell_1 - \ell_2}{\ell_2} \right) R'$$

where r = internal resistance

R' = Parallel resistance

ℓ_1 and ℓ_2 are null point length

7. **Ans. 60.00 m/sec**

$$\text{Sol. } v = \frac{2}{9} \frac{r^2}{\eta} (\rho_b - \rho_L) g$$

$$v \propto r^2$$

$$\frac{v_1}{v_2} = \frac{r_1^2}{r_2^2}$$

$$\frac{15}{v_2} = \frac{r^2}{4r^2}$$

$$v_2 = 60 \text{ m/sec}$$

8. **Ans. 6.40**

Sol. Consider an element of the slit of length dx at a depth x from the top.

Velocity of efflux at $x = \sqrt{2gx}$.

Force on elementary area = rate of change of momentum = $(b dx \rho v)v = b \rho dx (2gx)$

$\therefore F$ (total force experienced)

$$= 2b\rho g \int_{h-1}^h x dx = b\rho g \ell (2h - \ell)$$

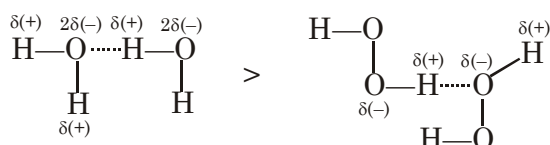
$$\text{or } F = 10^{-3} \times 1000 \times 10 \times 0.4 (2 - 0.4) = 6.40 \text{ N}$$

PART-2 : CHEMISTRY**SOLUTION****SECTION-I****1. Ans.(A,C,D)**

Sol. (A) As % s character in hybrid orbital increases electronegativity increases

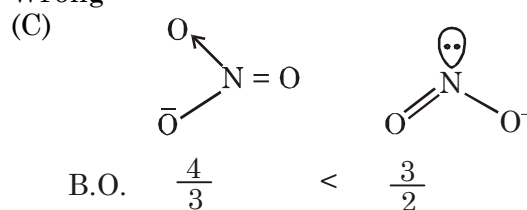
$$\begin{array}{ccc} sp & > & sp^2 > sp^3 \\ \% 's' & 50 & 33.3 \quad 25 \\ \text{correct} \end{array}$$

(B) The strength of hydrogen bonding depends on the polarity of H-A which further depends upon the difference in the electronegativity of H and A atoms



The accumulated negative charge on oxygen atom in H_2O is more as compared to that in H_2O_2 . So strength of hydrogen bonding in H_2O is more than H_2O_2 .

Wrong



d_{N-O} order $NO_3^- > NO_2^-$

correct

(D) Lattice energy increases with decrease in interionic distance (if charges are same)

Since charges are same here

Order is determined by interionic distance.

$BeO > MgO > CaO > SrO$

Correct

2. Ans.(A,D)**Sol.**

Complex	O.S.	Hybridisation	Magnetic nature	Geometry	C.No.
$[PtCl_4]^{2-}$	+2	dsp^2	Diamagnetic	Sq. planar	4
$[Pt(CN)_4]^{2-}$	+2	dsp^2	Diamagnetic	Sq. planar	4
$[Ni(CO)_4]$	0	sp^3	Diamagnetic	Tetrahedral	4

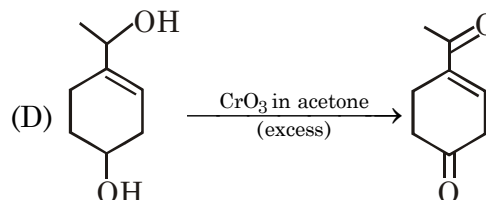
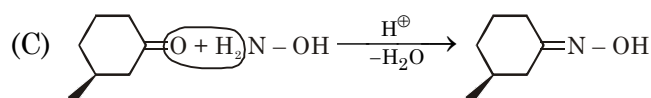
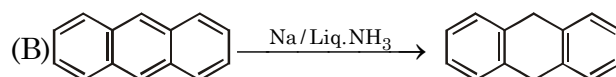
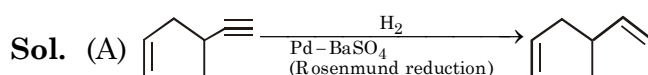
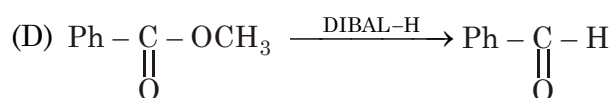
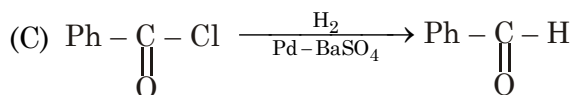
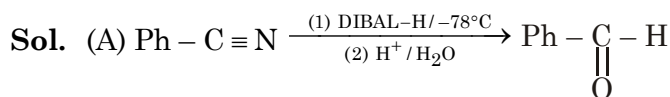
3. Ans.(A,D)

Sol. (i) Structure-(I) : This structure has no plane or axis of symmetry and this is unsymmetric and optically active.

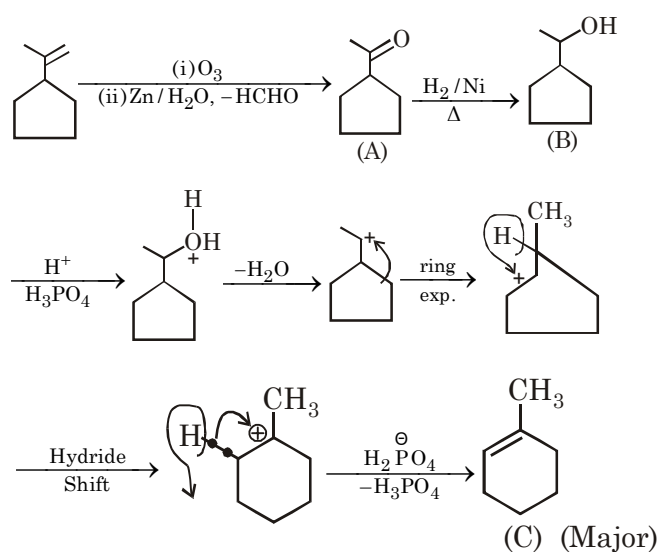
(ii) Structure-(II) : This structure has plane or axis of symmetry and this is symmetric and optically inactive

(iii) Structure-(III) : This structure has plane or axis of symmetry and this is symmetric and optically inactive.

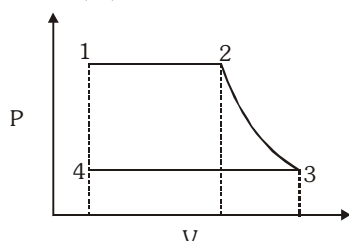
(iv) Structure-(IV) : This structure has no plane or axis of symmetry and this is unsymmetric and optically active.

4. Ans. (A,B,D)**5. Ans.(A,B,C,D)****6. Ans.(A,C,D)****7. Ans.(B)****8. Ans.(A)**

Solution for 7 & 8



9. Ans.(B)



Let initial volume is V litre

Volume at 600 K = 2V

$$\frac{T_3}{T_2} = \left(\frac{V_2}{V_3}\right)^{\gamma-1} = \left[\frac{2V}{4\sqrt{2}V}\right]^{\frac{2}{3}}$$

$$T_3 = T_2 \times \frac{1}{2} = 300 \text{ K}$$

$$\frac{V_3}{T_3} = \frac{V_4}{T_4} \Rightarrow \frac{4\sqrt{2}V}{300} = \frac{V}{T_4}$$

$$T_4 = \frac{300}{4\sqrt{2}} = \left[\frac{75}{\sqrt{2}}\right] \text{ K}$$

$$W = W_{12} + W_{23} + W_{34}$$

$$= -R(600 - 300) + \frac{3}{2}R(300 - 600) - R\left(\frac{75}{\sqrt{2}} - 300\right)$$

$$= -300R - \frac{900}{2}R + R300 - \frac{75}{\sqrt{2}}R$$

$$= -450R - \frac{75}{\sqrt{2}}R$$

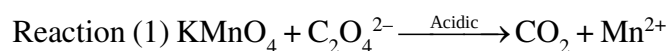
10. Ans.(A)

$$\Delta H = -\frac{5}{2}R(300) = -750 R$$

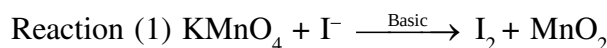
SECTION-II

1. Ans. 240.00

Let V_1 & V_2 are volumes (in ml) of KMnO_4 in 1st part & 2nd part



$$\Rightarrow 0.5 \times V_1 \times 5 = 1.5 \times 125 \times 2 \Rightarrow V_1 = 150 \text{ mL}$$



$$\Rightarrow 0.5 \times V_2 \times 3 = 0.5 \times 270 \times 1 \Rightarrow V_2 = 90 \text{ mL}$$

$$\text{So initial volume} = 150 + 90 = 240 \text{ mL}$$

2. Ans.(-450.00)

3. Ans.(-3.15)

4. Ans.(0.30)

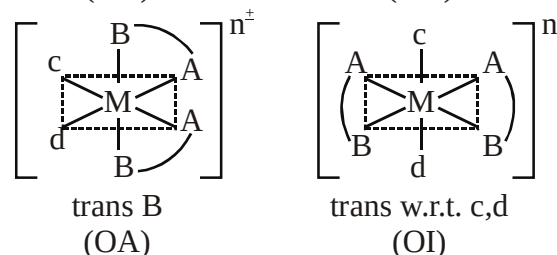
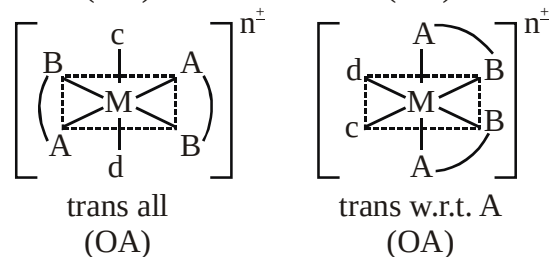
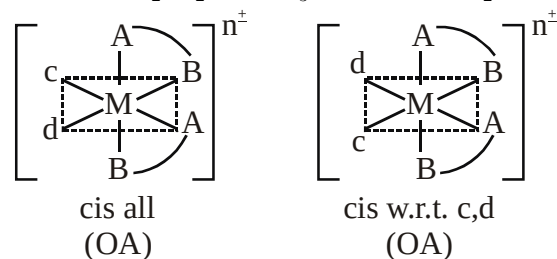
Sol. CO is carbonyl, if Z is atomic number of metal then, $Z + 5 \times 2 = 36 \Rightarrow Z = 26$

So, configuration is $[\text{Ar}] 3d^6, 4s^2$

$$\text{So ratio is } \frac{6}{20} = 0.30$$

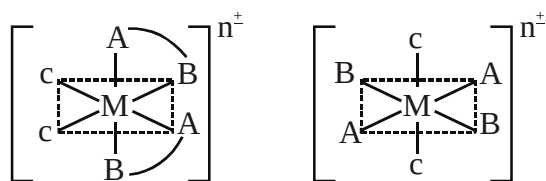
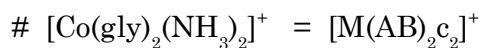
5. Ans.(1.50)

Sol. Case-I

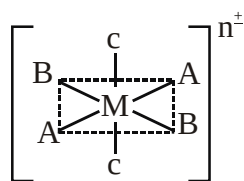


Total number of G.I. = 6

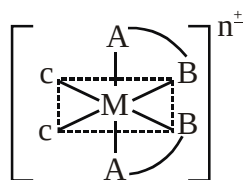
Total number of S.I. = $5 \times 2 + 1 = 11$



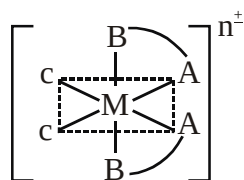
cis
(OA)



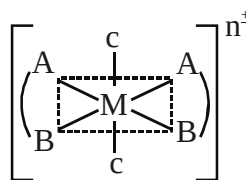
all trans
(OI)



trans w.r.t. A
(OA)



trans w.r.t. B
(OA)



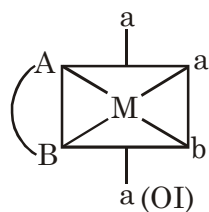
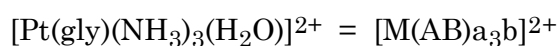
trans w.r.t. c
(OI)

Total number of G.I. = 5

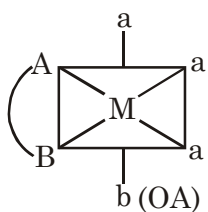
Total number of S.I. = $3 \times 2 + 2 = 8$

So, $X = 11 - 8 = 3$

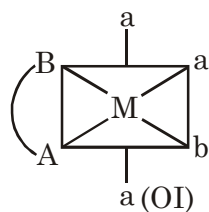
Case-II



a (OI)



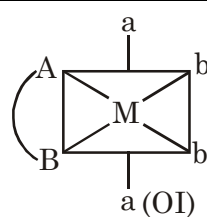
b (OA)



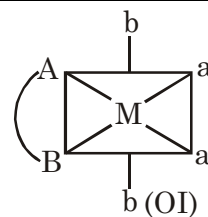
a (OI)

Total G.I. = 3

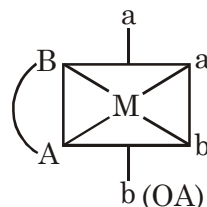
Total S.I. = $2 + 1 \times 2 = 4$



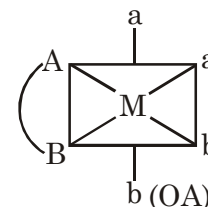
a (OI)



b (OI)



b (OA)



b (OA)

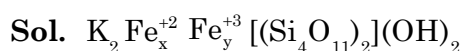
Total G.I. = 4

Total S.I. = $2 + 2 \times 2 = 6$

$Y = 6 - 4 = 2$

$$\text{So } \frac{x}{y} = \frac{3}{2} = 1.50$$

6. Ans.(1.50)



By charge balance

$$2 + 2x + 3y + (4 \times 4 - 22) \times 2 - 2 = 0 \quad (1)$$

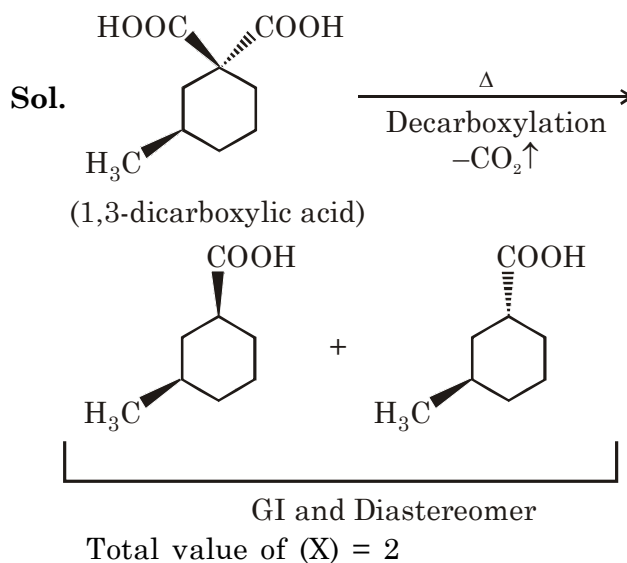
$$x + y = 5 \quad (2)$$

on solving 'x' and 'y'

$$x = 3, y = 2$$

$$\text{so ratio } \frac{3}{2} = 1.5$$

7. Ans.(2.00)

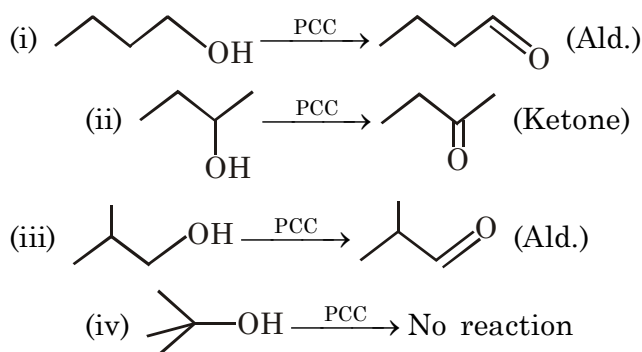


8. Ans. (3.00)

Sol. $C_4H_{10}O \rightarrow$

DBE = O $\begin{cases} \text{Ether (Not react with PCC)} \\ \text{Alcohol (Oxidise with PCC)} \end{cases}$

Possible alcohol



PART-3 : MATHEMATICS

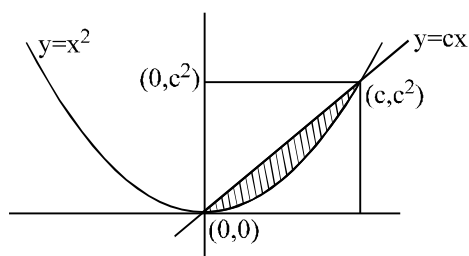
SOLUTION

SECTION-I

1. Ans. (A,C)

Sol. Area (T) = $\frac{c \cdot c^2}{2} = \frac{c^3}{2}$

$$\text{Area (R)} = \frac{c^3}{2} - \int_0^c x^2 dx$$



$$= \frac{c^3}{2} - \frac{c^3}{3} = \frac{c^3}{6}$$

$$\therefore \lim_{c \rightarrow 0^+} \frac{\text{Area (T)}}{\text{Area (R)}} = \lim_{c \rightarrow 0^+} \frac{c^3}{2} \cdot \frac{6}{c^3} = 3$$

2. Ans. (C)

Sol. $AB = \begin{bmatrix} -x & -y & 7z \\ 0 & y & 14z \\ x & -y & 7z \end{bmatrix}$

equating coefficients $x^2 = 4$; $y^2 = 9$
and $z^2 = 1$

$$\Rightarrow x = \pm 2, y = \pm 3 \text{ and } z = \pm 1$$

number of solution = $2 \times 2 \times 2 = 8$ Ans.

3. Ans. (A,D)

Sol. $(2x - y - 5)\vec{a} = (x - 2y - 4)\vec{b}$

$$\therefore 2x - y = 5 \quad \dots(1)$$

$$x - 2y = 4 \quad \dots(2)$$

from (1) and (2)

$$2(2y + 4) - y = 5$$

$$3y = -3 \Rightarrow y = -1$$

$$\text{and } x = 2$$

hence $x + y = 1$ Ans.

4. Ans. (A,B,C,D)

Sol. (A) $AB = 0$

pre multiply A^{-1}

$$A^{-1}AB = A^{-1} \times 0$$

$$B = 0$$

(B) similarly

(C) Assume A as non singular

$$AB = 0$$

pre multiply with A^{-1}

$$A^{-1}AB = A^{-1} \cdot 0$$

$B = 0$ which is contradiction

Hence A must be singular

(at least one of A or B)

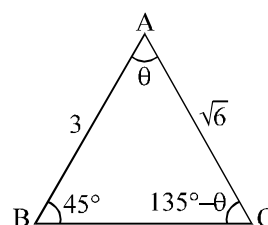
(D) $AB = 0$

$$|AB| = |0| \Rightarrow |A| = 0 \text{ or } |B| = 0$$

5. Ans. (C)

Sol. $\frac{\sin(135^\circ - \theta)}{3} = \frac{\sin 45^\circ}{\sqrt{6}}$

$$\sin(135^\circ - \theta) = \frac{\sqrt{3}}{2}$$



$$135^\circ - \theta = 60^\circ \text{ or } 120^\circ$$

$$\theta = 75^\circ \text{ or } 15^\circ$$

$\theta = 15^\circ$ (not possible as the Δ is acute)

$$\therefore \theta = 75^\circ \quad \text{Ans.}$$

6. Ans. (A,B,C)

Sol. $A^2 = A$

$$(A^2)^T = A^T \Rightarrow (A^T)^2 = A^T \Rightarrow (-A)^2 = -A$$

$$A^2 = -A \Rightarrow A = -A \Rightarrow A = 0$$

Paragraph for Question 7 and 8

Sol. Note that C_1 is a semicircle and C_2 is a rectangular hyperbola.

PQ will be minimum if the normal at P on the semicircle is also a normal at Q on $xy = 9$

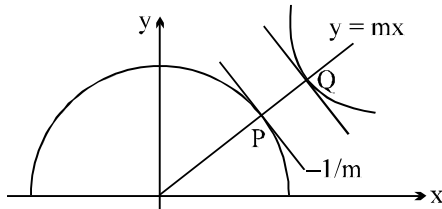
Let the normal at P be $y = mx$ (1)

($m > 0$)

solving it with

$$xy = 9$$

$$mx^2 = 9 \Rightarrow x = \frac{3}{\sqrt{m}}; y = \frac{9\sqrt{m}}{3}$$



$$\therefore Q \equiv \left(\frac{3}{\sqrt{m}}, 3\sqrt{m} \right)$$

differentiating $xy = 9$

$$x \frac{dy}{dx} + y = 0 \Rightarrow \frac{dy}{dx} = -\frac{y}{x}$$

$$\therefore \left. \frac{dy}{dx} \right|_Q = -\frac{3\sqrt{m} \cdot \sqrt{m}}{3} = -m$$

$\therefore$ tangent at P and Q must be parallel

$$\therefore -m = -\frac{1}{m} \Rightarrow m^2 = 1 \Rightarrow m = 1$$

$\therefore$ normal at P and Q is $y = x$

solving P(1, 1) and Q(3, 3)

$$\therefore (PQ)^2 = d^2 = 4 + 4 = 8 \text{ Ans.}$$

7. Ans. (B)

8. Ans. (C)

Paragraph for Question 9 and 10

$$\text{Sol. } x \int_0^x g(t) dt + \int_0^x (1-t) g(t) dt = x^4 + x^2$$

differentiate w.r.t. 'x'

$$xg(x) + \int_0^x g(t) dt + (1-x)g(x) = 4x^3 + 2x \dots(1)$$

9. Ans. (C)

Sol. From (1)

$$\int_0^x g(t) dt + g(x) = 4x^3 + 2x$$

$$\text{let } g(x) + g'(x) = 12x^2 + 2 \Rightarrow \frac{dy}{dx} + y = 12x^2 + 2$$

$$(\because y = g(x))$$

10. Ans. (A)

Put $x = 0$ in (1) we get $g(0) = 0$

SECTION-II

1. Ans. 1.00

Sol. $f'(x) = 3x^2 + 2 > 0, \forall x \in \mathbb{R}$

$\Rightarrow$ increasing ($\uparrow$) $\Rightarrow$ (B)

2. Ans. 0.50

$$\text{Sol. } \lim_{x \rightarrow \infty} \frac{x^{3/2}}{(x-1)} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2[1-(1/x)]} = \frac{1}{2} \text{ Ans.}$$

3. Ans. -16.00

$$\text{Sol. } \det(B) = \begin{vmatrix} 4x & 2a & -p \\ 4y & 2b & -q \\ 4z & 2c & -r \end{vmatrix}$$

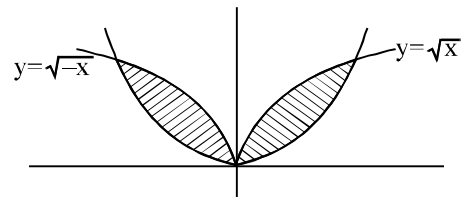
$$= (4)(2)(-1) \begin{vmatrix} x & a & p \\ y & b & q \\ z & c & r \end{vmatrix}$$

$$= -8 \begin{vmatrix} x & y & z \\ a & b & c \\ p & q & r \end{vmatrix} = -8 \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$$

4. Ans. 0.66 or 0.67

$$\text{Sol. } A = \left(\frac{16ab}{3} \right) \cdot 2$$

$$a = \frac{1}{4}; b = \frac{1}{4}$$



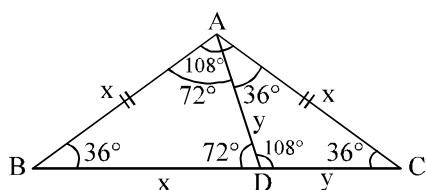
$$A = \frac{2}{3} \text{ Ans.}$$

5. Ans. 1.61 or 1.62

Sol. In ΔABC $\left(\frac{x}{y} = ?\right)$

$$\frac{x+y}{\sin 108^\circ} = \frac{x}{\sin 36^\circ}$$

$$\frac{x+y}{x} = \frac{\sin 108^\circ}{\sin 36^\circ} \quad \dots(1)$$



$$1 + \frac{y}{x} = \frac{\sin 72^\circ}{\sin 36^\circ} = 2 \cos 36^\circ$$

$$\frac{y}{x} = \frac{\sqrt{5}+1}{2} - 1 = \frac{\sqrt{5}-1}{2}$$

$$\frac{x}{y} = \frac{2(\sqrt{5}+1)}{(\sqrt{5}-1)(\sqrt{5}+1)} = \frac{\sqrt{5}+1}{2} \text{ Ans.}$$

6. Ans. 2.00

Sol. $\frac{dy}{dx} = xy(1+y)$

$$\int \frac{dy}{(1+y)y} = \int x \cdot dx$$

$$\frac{2y}{1+y} = e^{\frac{x^2}{2}} \quad (\because f(0) = 1)$$

$$f(2) = \frac{e^2}{2 - e^2}$$

7. Ans. 9.00

Sol. $y^2 = \alpha x^3 - \beta$

if (2, 3) is lie on the curve

$$8\alpha - \beta = 9 \quad \dots(1)$$

slope of normal at (2, 3)

$$-\frac{1}{4} = -\frac{1}{2\alpha} \Rightarrow \alpha = 2$$

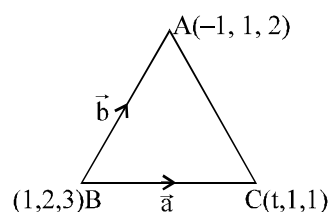
8. Ans. 2007.00

Sol. $A = \frac{1}{2} |\vec{a} \times \vec{b}|$ and

$$|\vec{a} \times \vec{b}|^2 = \vec{a}^2 \vec{b}^2 - (\vec{a} \cdot \vec{b})^2$$

$$\vec{a} = (t-1)\hat{i} - \hat{j} - 2\hat{k}; \quad \vec{b} = 2\hat{i} + \hat{j} + \hat{k}$$

$$|\vec{a}|^2 = (t-1)^2 + 1 + 4; \quad |\vec{b}|^2 = 4 + 1 + 1 = 6$$



$$\vec{a} \cdot \vec{b} = 2(t-1) - 1 - 2 = 2t - 5$$

$$|\vec{a} \times \vec{b}|^2 = 6[t^2 - 2t + 6] - (4t^2 + 25 - 20t)$$

$$|\vec{a} \times \vec{b}|^2 = 2t^2 + 8t + 11$$

which is minimum at $t = -2$

$$|\vec{a} \times \vec{b}|_{\min}^2 = 8 - 16 + 11 = 3$$

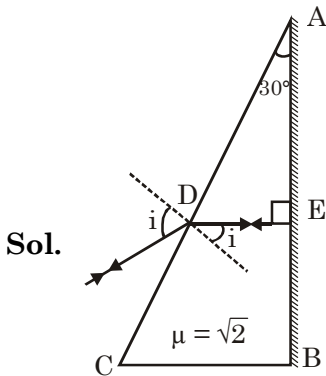
$$|\vec{a} \times \vec{b}|_{\min} = \sqrt{3}$$

$$\therefore \frac{|\vec{a} \times \vec{b}|_{\min}}{2} = A_{\min} = \frac{\sqrt{3}}{2}$$

$$\text{hence } (1338\sqrt{3}) \cdot \frac{\sqrt{3}}{2} = 669 \times 3 = 2007 \text{ Ans.}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (A)



As incident ray retraces its path the ray is incident normally on the silver face of the prism as shown in figure.

Further, as in $\triangle AED$ $30^\circ + 90^\circ + \angle D = 180^\circ$
 $\Rightarrow \angle D = 60^\circ$

Now as by construction, $\angle D + \angle r = 90^\circ$
 $\Rightarrow \angle r = 90^\circ - 60^\circ = 30^\circ$

$\therefore$ from Snell's law at surface AC,

$$1 \sin i = \sqrt{3} \sin 30^\circ = \sqrt{3} \times \frac{1}{2} = \frac{\sqrt{3}}{2}$$

$$\therefore \sin i = \frac{\sqrt{3}}{2} \text{ so } i = 60^\circ$$

2. Ans. (B)

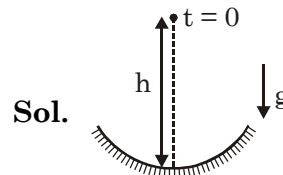
Sol. If D is the distance between the object and the screen, d the separation of the two position of lens throwing two images on the screen then

$$m_1 = \frac{(D+d)}{(D-d)} \text{ and } m_2 = \frac{(D-d)}{(D+d)}$$

$$\therefore m_1 - m_2 = \frac{4Dd}{D^2 - d^2} \text{ but } \frac{D^2 - d^2}{4D} = f$$

$$\text{so } m_1 - m_2 = \frac{d}{f} \Rightarrow f = \frac{d}{m_1 - m_2}$$

3. Ans. (C)

**Sol.**

$$v_{\text{image}} = -m^2 v_{\text{object}} = -m^2(gt) \text{ where}$$

$$m = \frac{f}{f-u} = \frac{-f}{-f + \left(\frac{f}{2} - \frac{gt^2}{2}\right)} = \frac{2f}{f+gt^2}$$

$$\Rightarrow v_1 = -\left(\frac{2f}{f+gt^2}\right)^2 (gt) = \frac{-4t^2gt}{(f+gt^2)^2}$$

$$\text{For maximum speed } \frac{dv_1}{dt} = 0 \Rightarrow t = \sqrt{\frac{f}{3g}}$$

$$\Rightarrow v_{\text{max}} = \frac{3}{4}\sqrt{3fg} = 22.5 \text{ cm/s}$$

4. Ans. (C)

5. Ans. (B)

Sol. friction force becomes zero, when normal contact force become zero. If v is the speed of particle at that moment, then

$$\frac{mv^2}{(R/2)} = qvB \Rightarrow v = \frac{qBR}{2m}$$

$\therefore$ change in K.E.

$$\Delta K = \frac{1}{2}m\left(\frac{qBR}{m}\right)^2 - \frac{1}{2}m\left(\frac{qBR}{2m}\right)^2$$

$$\Delta K = \frac{1}{2} \frac{q^2 B^2 R^2}{m} \left[1 - \frac{1}{4}\right]$$

$$\therefore \text{work done by friction} = -\frac{3}{8} \frac{q^2 B^2 R^2}{m}$$

$$\therefore K = 3$$

6. Ans. (C)

$$\text{Sol. } \vec{F} = I(\vec{\ell} \times \vec{B}) = 4(8\hat{i} \times 5\hat{k}) = -160\hat{j} \text{ N/}$$

7. Ans. (B)

Sol. $q = v \cos \theta B = Mg$

$$B = \frac{Mg}{qv \cos \theta}$$

8. Ans. (D)

Sol. For force on q to be zero, charges q and Q must be of opposite of nature. Net attraction force on q due to charges

Q = Repulsion force on q due to $q\sqrt{2}$

$$\text{so } \sqrt{2} F_A = F_R$$

$$\Rightarrow \sqrt{2} \frac{kQq}{a^2} = \frac{kq^2\sqrt{2}}{(\sqrt{2}a)^2} \Rightarrow |q| = 2Q$$

Hence $q = -2Q$

$$Q = -\frac{q}{2}$$

9. Ans. (C)

Sol. $B_0 = B_1 - B_2 + B_3 - B_4 + B_5 \dots \dots \alpha$

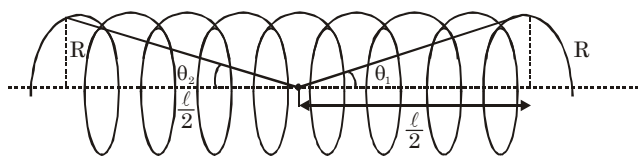
$$= \frac{\mu_0 i \theta}{4\pi a} - \frac{\mu_0 i (\theta)}{4\pi (2a)} + \frac{\mu_0 i (\theta)}{4\pi (3a)} - \frac{\mu_0 i \theta}{4\pi (4a)} + \dots \dots \infty$$

$$= \frac{\mu_0 i (\theta)}{4\pi a} \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \dots \infty \right)$$

$$= \frac{\mu_0 i (\theta)}{4\pi a} (\ln 2)$$

10. Ans. (B)

Sol. $B_{\text{axis}} = \frac{\mu_0 n i}{2} (\cos \theta_1 + \cos \theta_2)$



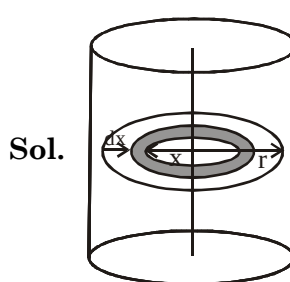
$$\tan \theta_1 = \frac{2R}{\ell} \Rightarrow \cos \theta_1 = \frac{\ell}{\sqrt{4R^2 + \ell^2}}$$

due to symmetry, $\theta_2 = \theta_1$

$$\cos \theta_2 = \cos \theta_1$$

$$B_{\text{center}} = \frac{\mu_0 n i}{2} (2 \cos \theta_1) = \frac{\mu_0 n i \ell}{\sqrt{4R^2 + \ell^2}}$$

11. Ans. (D)



Sol.

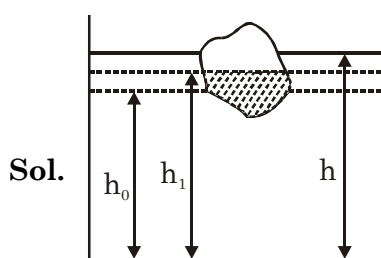
$$\oint B d\ell = \mu_0 I_{\text{en}}$$

$$\oint B_0 r^3 d\ell = \mu_0 I_{\text{en}}$$

or $B_0 r^3 \int_0^{2\pi} d\ell = \mu_0 I_{\text{en}}$ [Note : $B_0 r^3$ is taken out of integration because it is not varying along $d\ell$]

$$\text{or } B_0 r^3 (2\pi r) = \mu_0 I_{\text{en}} \quad \text{or } I_{\text{en}} = \frac{2\pi B_0 r^4}{\mu_0}$$

12. Ans. (B)



Sol.

Water formed as a result of melting ice has a volume corresponding to the hatched region in the figure. Since a part of this volume is above the surface of water ($h_1 > h_0$), the level of water rises after melting of ice. On the other hand, since $h_1 < h$, oil fills the formed "hole", i.e. the total level of liquid in the vessel falls.

13. Ans. (B)

Sol. $\ell = 15 \text{ cm} = 15 \times 10^{-2} \text{ m}$;
 $a = 1.4 \text{ cm} = 1.4 \times 10^{-2} \text{ m}$;
 $b = 1.5 \text{ cm} = 1.5 \times 10^{-2} \text{ m}$;
 $q = 3.5 \mu\text{C} = 3.5 \times 10^{-6} \text{ C}$

$$\text{Capacitance } C = \frac{2\pi\epsilon_0 \ell}{\ln\left(\frac{b}{a}\right)}$$

$$= \frac{2\pi \times 8.854 \times 10^{-12} \times 15 \times 10^{-2}}{\ln\left(\frac{1.5 \times 10^{-2}}{1.4 \times 10^{-2}}\right)} = 1.2 \times 10^{-10} \text{ F}$$

Since the outer cylinder is earthed, the potential of the inner cylinder will be equal to the potential difference between them. Potential of inner cylinder, is

$$V = \frac{q}{C} = \frac{3.5 \times 10^{-6}}{1.2 \times 10^{-10}} = 2.9 \times 10^4 \text{ V}$$

14. **Ans. (D)**

Sol. Since collision is elastic $\therefore$ Velocity of dielectric after collision is $\frac{2v_0}{3}$

Dielectric will move and when it is coming out of capacitor a force is applied on it by

$$\text{the capacitor } F = -\frac{dU}{dx} = -\frac{E^2 \epsilon_0 b (K-1)}{2d}$$

Which decreases its speed to zero, till it comes out it travels a distance a

$$\frac{4}{9} \times 2 \times \frac{1}{2} M v_0^2 = \frac{E^2 \epsilon_0 b (K-1) a}{2d}$$

$$v_0 = E \left[\frac{\epsilon_0 a b (K-1)}{2 M d} \right]^{1/2} \times \frac{3}{2}$$

15. **Ans. (C)**

Sol. $Mg = mg + B$

$$Mg = mg + p_2 \times \frac{M}{p_1} g$$

$$M \left(1 - \frac{p_2}{p_1} \right) = m = 6 \text{ kg}$$

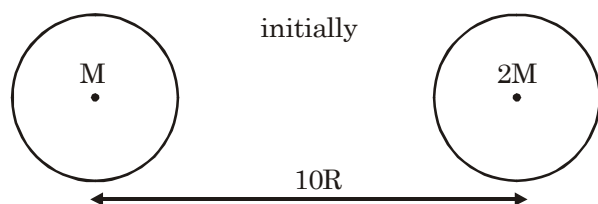
16. **Ans. (B)**

Sol. $BI \ell \frac{\ell}{2} = \frac{mg}{2} \ell \cos \theta$

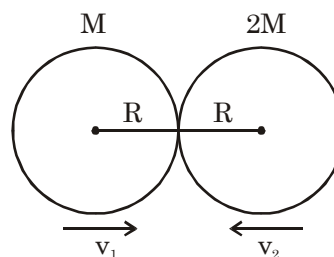
$$\cos \theta = \frac{BI \ell}{mg}$$

$$= \frac{0.36 \times 4.0 \times 0.45}{0.09 \times 10} = 0.72$$

17. **Ans. (A)**



$$U_i = -\frac{GM(2m)}{10R} \text{ \& } K_i = 0$$



$$U_f = -\frac{GM(2M)}{2R} \text{ \& } K_f = \frac{1}{2} M V_1^2 + \frac{1}{2} 2 M V_2^2$$

Conservation of energy :

$$U_i + K_i = U_f + K_f$$

$$-\frac{GM(2M)}{10R} + 0 = -\frac{GM(2M)}{2R} + \frac{1}{2} M V_1^2 + \frac{1}{2} 2 M V_2^2$$

also, conservation of momentum

$$0 = M V_1 + 2 M (-V_2)$$

$$V_1 = 2 V_2$$

Solving the above two equation

$$v_1 = \sqrt{\frac{16GM}{15R}} = 4 \sqrt{\frac{GM}{15R}}$$

$$\text{\& } v_2 = \frac{v_1}{2} = \sqrt{\frac{4GM}{15R}} = 2 \sqrt{\frac{GM}{15R}}$$

$$v_{\text{rel}} = v_1 + v_2 = 6 \sqrt{\frac{GM}{15R}} \text{ Answer.}$$

18. **Ans. (C)**

19. **Ans. (C)**

Sol. The wire's resistance is

$$R = \frac{\rho L}{A} = \frac{\rho L}{\pi r^2} = 3.8 \Omega$$

. Thus the current in the wire is

$$\ell = \frac{\Delta V_{\text{min}}}{R} = \frac{1.5 \text{ V}}{3.8 \Omega} = 0.39 \text{ A}$$

20. **Ans. (A)**

Sol. $I = n c A v_d$

$$v_d \propto \frac{1}{A} \propto \frac{1}{r^2}$$

$$\frac{v_{d\text{left}}}{v_{d\text{right}}} = \frac{r_{\text{right}}^2}{r_{\text{left}}^2} = \left(\frac{1}{3} \right)^2$$

$$v_{d\text{left}} = \frac{v_{d\text{right}}}{9} = \frac{0.50 \times 10^{-4}}{9} \text{ m/s} = 0.055 \text{ m/s}$$

SECTION-II

1. Ans. 2.50

Sol. The mean time between collision is related to a material's conductivity by

$$\tau = \frac{m\sigma}{n_e e^2}$$

$$\tau = \frac{(9.1 \times 10^{-31} \text{ kg})(6.0 \times 10^7 \Omega^{-1} \text{ m}^{-1})}{(8.5 \times 10^{28} \text{ m}^{-3})(1.60 \times 10^{-19} \text{ C})^2}$$

$$= 2.5 \times 10^{-14} \text{ s}$$

2. Ans. 0.64

Sol. Energy losses $\Delta U = \frac{1}{2} CV_0^2 - \frac{1}{2} CV^2$
Fractional energy loss

$$\frac{\Delta U}{U_0} = \frac{\frac{1}{2} CV_0^2 - \frac{1}{2} CV^2}{\frac{1}{2} CV_0^2} = \frac{V_0^2 - V^2}{V_0^2} = \frac{(100)^2 - (60)^2}{(100)^2}$$

$$= 0.64$$

3. Ans. 0.75

Sol. Let the equivalent capacitance of the combination = C_{eq}

$$\frac{1}{C_{eq}} = \frac{1}{C} + \frac{1}{4C} + \frac{1}{16C} + \dots \infty$$

$$= \left[1 + \frac{1}{4} + \frac{1}{16} + \dots \infty \right] \frac{1}{C} \text{ (this is G.P. series)}$$

$$\frac{1}{C_{eq}} = \left(\frac{1}{1 - \frac{1}{4}} \right) \times \frac{1}{C}$$

$$\Rightarrow C_{eq} = \frac{3C}{4}$$

$$\Rightarrow \frac{C_{eq}}{C} = \frac{3}{4} = 0.75$$

4. Ans. 6.00

Sol. $mg = [I 2\pi R B] \sin 30^\circ$

5. Ans. 138.39

Sol. $\vec{E} = -\frac{\partial v}{\partial x} \hat{i} - \frac{\partial v}{\partial y} \hat{j} - \frac{\partial v}{\partial z} \hat{k}$
 $= -(y^2 z^3) \hat{i} - (x(2y)z^3) \hat{j} - (xy^2(3z^2)) \hat{k}$
at (1, -3, 2)

$$\vec{E} = -(72) \hat{i} - (-48) \hat{j} - (108) \hat{k}$$

$$\vec{E} = -72 \hat{i} + 48 \hat{j} - 108 \hat{k}$$

$$|E| = 138.39 \text{ v/m}$$

PART-2 : CHEMISTRY

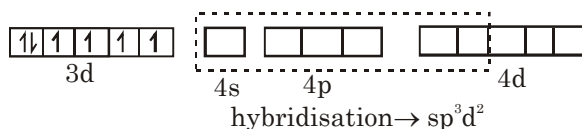
SOLUTION

SECTION-I

1. Ans.(C)

Sol. $\text{Co}^{3+} : [\text{Ar}] 3d^6$

This complex is outer orbital complex



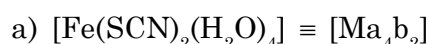
$$n = 4 \quad \mu = \sqrt{4(4+2)} \approx 4.9 \text{ B.M.}$$

Geometry : octahedral

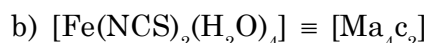
2. Ans.(A)

Sol. (A) $[\text{Fe}(\text{SCN})_2(\text{H}_2\text{O})_4]$

SCN^- is ambidentate ligand so, three linkage isomers

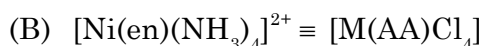


$$\text{G.I.} = 2$$

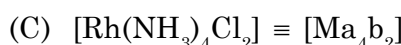


$$\text{G.I.} = 2$$

$$\text{So total isomers are} = 3 + 2 + 2 + 2 = 9$$



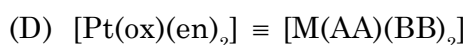
does not show structural as well as stereo isomerism



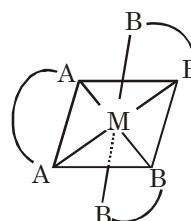
Does not show structural isomerism

But show stereo (only geometrical) isomerism two geometrical isomers

So total isomers are : 2



does not show structural isomerism show stereo isomerism



No GI
optically active
stereo isomers = 2
total isomers = 2

3. Ans.(A)

Sol. When number of ions are same the solution would show nearly same electrical conductance.

	number of ions
(A) $[\text{Pd}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$	: 4
$[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$	: 4
(B) $[\text{Pd}(\text{NH}_3)_6]\text{Cl}_4$	: 5
$[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$	: 3
(C) $[\text{Pd}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$	: 4
$[\text{CrCl}_3(\text{NH}_3)_3]$	: 0
(D) $[\text{Pd}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$	: 3
$[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$	: 2

4. Ans.(C)

Sol. In isoelectronic metal carbonyl as -ve charge on metal increase M-C bond order increases.

	Oxidation state of metal
(A) $[\text{Ni}(\text{CO})_4]$	0
(B) $[\text{Fe}(\text{CO})_5]$	0
(C) $[\text{V}(\text{CO})_6]^-$	-1
(D) $[\text{Mn}(\text{CO})_6]^+$	+1

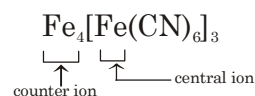
So answer is (C)

5. Ans.(C)

Sol. Fe^{3+} : Ferri ion

Fe^{2+} : Ferro ion

So, Formula is



So, O.S. of counter 'Fe' i.e. c = +3

O.S. of central 'Fe' i.e. b = +2

$$\text{Avg O.S.} = \frac{4 \times 3 + 3 \times 2}{7} = +\frac{18}{7}$$

6. Ans.(C)

Sol. (A) Polarising power of cation depends on ionic potential of cation (ϕ).

As ϕ increases polarising power increases

$$\phi = \frac{\text{charge}}{\text{radius}}$$

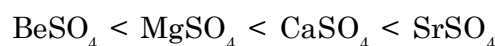
$\therefore$ charges are same, but size of Be^{2+} is

smallest and Ba^{2+} is largest so order should be $\text{Be}^{2+} > \text{Mg}^{2+} > \text{Ca}^{2+} > \text{Ba}^{2+}$

(B) For s-block thermal stability in case of polyatomic anion is decided by covalent nature. As covalent nature increases thermal stability decreases.

BeSO_4 is having maximum covalent nature so least thermal stable

order should be



(C) Due to $2p_\pi - 2p_\pi$ effective back bonding in BF_3 it lewis acidic nature is least. So given order is correct.

$$\begin{array}{ccc} \text{O}_2 & \text{O}_2^- & \text{O}_2^{2-} \\ \text{(D) B.O.} & \frac{10-6}{2} & \frac{10-7}{2} & \frac{10-8}{2} \\ & = 2.0 & = 1.5 & = 1.0 \end{array}$$

So, order should be $\text{O}_2 > \text{KO}_2 > \text{K}_2\text{O}_2$

7. Ans.(B)

Sol. (A) All four quantum number are given so only for one e^-

(B) $\hat{n} = 5 \Rightarrow$ means five subshell and each subshell has one orbital for $m = 0$

Hence total orbitals are five which can accomodate maximum ten electrons out of

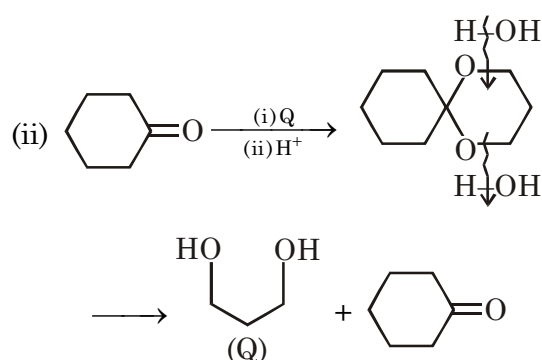
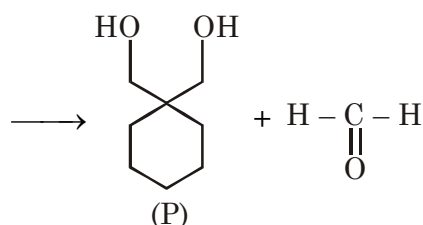
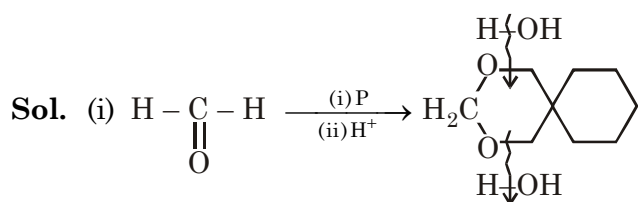
which five electrons would have spin $+\frac{1}{2}$

(C) It represents one '6d' orbital i.e. for two electrons

(D) $\hat{n} = 3 \Rightarrow$ means three subshell and each subshell has one orbital for $m = 0$

Hence total orbitals are three which can accomodate maximum six electrons out of which three electrons would have spin $+\frac{1}{2}$.

8. Ans.(B)



9. Ans.(C)

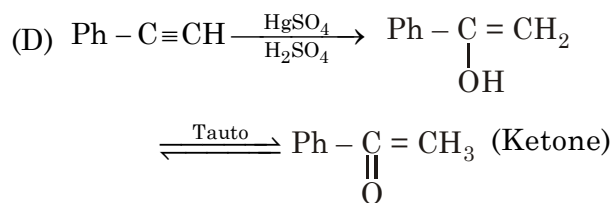
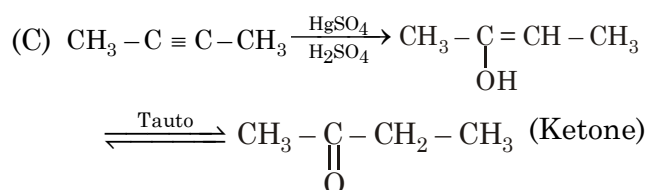
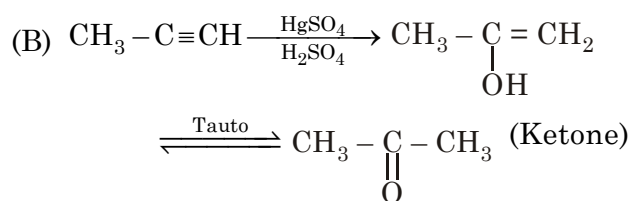
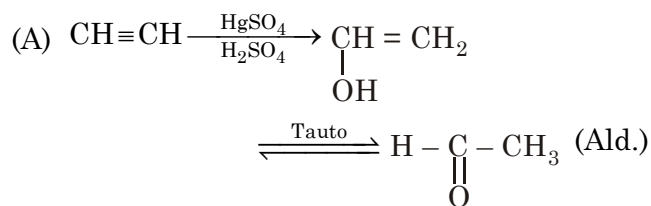
Sol.

Rate of NAR $\propto \delta$ positive on carbonyl carbon

$$\propto \frac{1}{\text{Steric hindrance factor}}$$

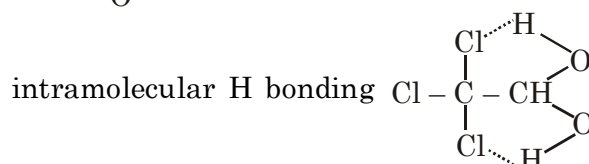
10. Ans.(A)

Sol.



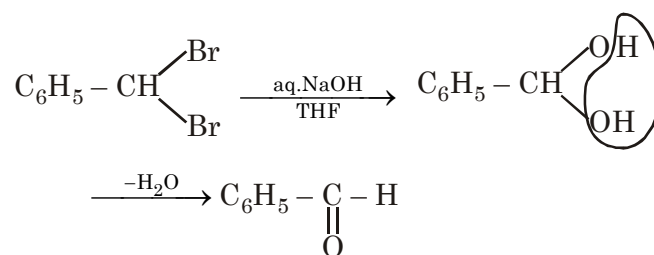
11. Ans.(B)

Sol. $\text{CCl}_3-\text{C}(=\text{O})-\text{H}$ from stable hydrate due to

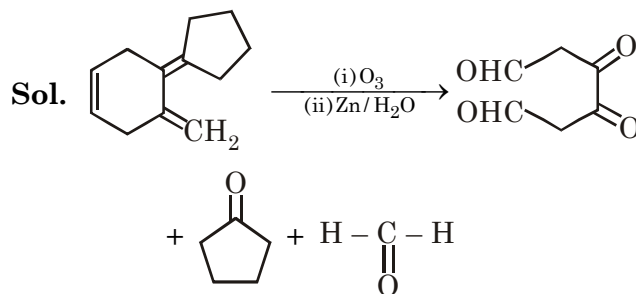


12. Ans.(B)

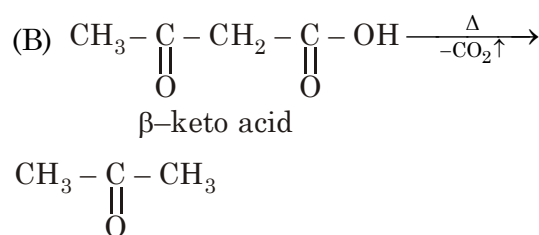
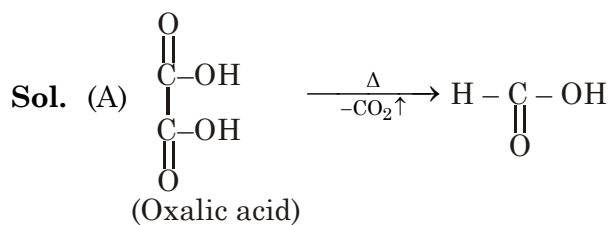
Sol.

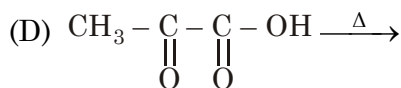
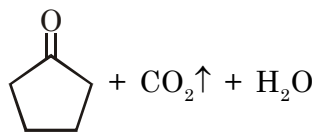
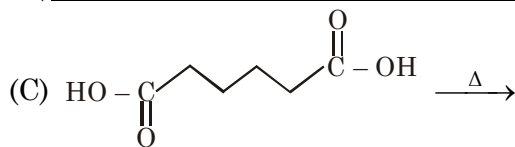


13. Ans.(C)



14. Ans.(D)





No reaction, No CO_2 release

15. Ans. (C)

16. Ans. (C)

$$\begin{aligned} T_b - T_f &= 105 \Rightarrow \Delta T_b + 100 - (-\Delta T_f) = 105 \\ &\Rightarrow \Delta T_b + \Delta T_f = 5 \\ &\Rightarrow 2.5m = 5 \Rightarrow m = 2 \end{aligned}$$

$$m = \frac{W \times 1000}{180 \times 100} \Rightarrow W = 36 \text{ gm}$$

17. Ans. (D)

Sol. $k = \frac{1}{t} \ln \left(\frac{V_0 - V_\infty}{V_t - V_\infty} \right) \Rightarrow \frac{1}{10} \ln \left(\frac{6}{5} \right) = 0.02 \text{ min}^{-1}$

18. Ans. (A)

$$\frac{A_\alpha^0}{A_\beta^0} = \frac{8}{1} \left| \frac{A_\alpha}{A_\beta} \right| = \frac{A_\alpha^0 e^{-\gamma_\alpha \times 2}}{A_\beta^0 e^{-\gamma_\beta \times 2}} \Rightarrow \frac{1}{1} = \frac{1}{8} e^{(\lambda_\beta - \lambda_\alpha) \times 2}$$

$$\Rightarrow (2)^3 = e^{(\lambda_\beta - \lambda_\alpha) \times 2 \times 60}$$

$$\Rightarrow 3(\ln 2) = \left(\frac{\ln 2}{20} - \frac{\ln 2}{t_{1/2}} \right) 120$$

$$t_{1/2} = 40 \text{ min.}$$

19. Ans. (B)

$$W_{AB} (\text{isochoric}) = 0, W_{BC} = -nRT \ln (P_1/P_2) \quad (\text{isothermal})$$

$$= -1 \times 0.08 \times 400 \ln 2$$

$$= -0.08 \times 400 \times 0.7 \Rightarrow -22.4 \text{ litre atm}$$

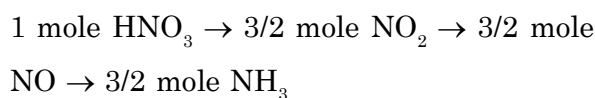
$$W_{AB} (\text{isobaric}) = -nR\Delta T \Rightarrow -1 \times 0.08 (200 - 400)$$

$$= 16 \text{ litre atm}$$

$$W_{\text{Total}} = W_{AB} + W_{BC} + W_{CA} \Rightarrow 0 - 22.4 + 16$$

$$\Rightarrow 6.4 \text{ litre atm}$$

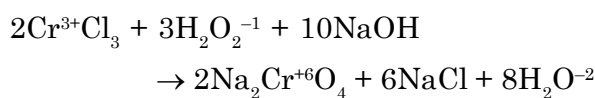
20. Ans. (C)



$$\Delta H_T = \left(-\frac{1}{4} \times \frac{3}{2} \times 200 \right) + \left(-\frac{1}{2} \times \frac{3}{2} \times 80 \right) + \left(-\frac{1}{2} \times 100 \right) = 185$$

SECTION-II

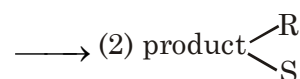
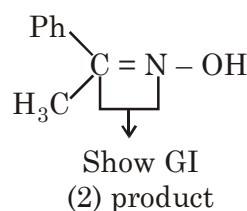
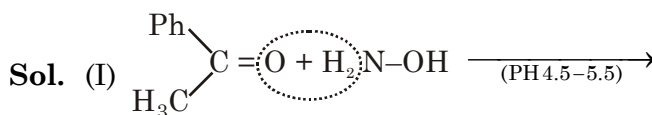
1. Ans. (15.00)



$$a = 2, b = 3, c = 10$$

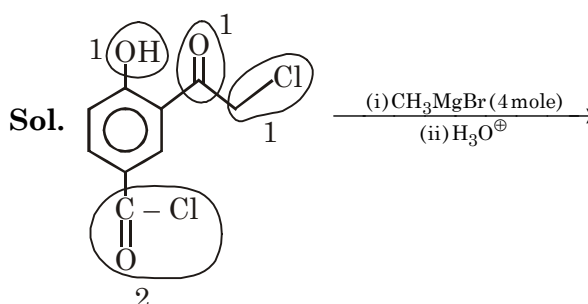
$$2 + 3 + 10 = 15$$

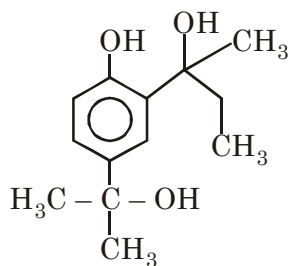
2. Ans. (4.00)



$$\text{Total } \begin{matrix} x=2 \\ y=2 \end{matrix} \left\{ \text{total (4) products} \right.$$

3. Ans. (5.00)



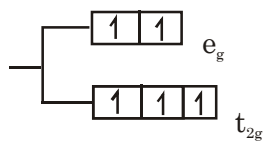


4. Ans.(0.00)

Sol. $\mu = \sqrt{35}$

$\Rightarrow n = 5$ (no. of unpaired e^-)

Fe^{3+} in octahedral field (for $n = 5$)

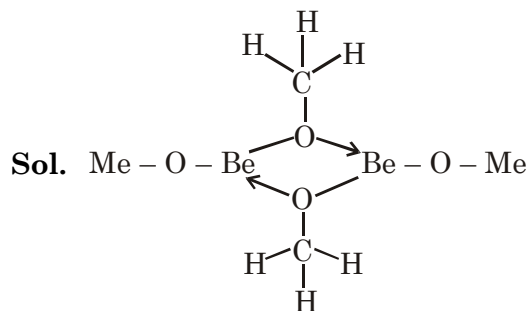


$$\text{C.F.S.E.} = -0.4\Delta_0 n_{t_{2g}} + 0.6\Delta_0 n_{e_g} + mp$$

$$= -0.4\Delta_0 \times 3 + 0.6\Delta_0 \times 2 + 0$$

$$= 0$$

5. Ans.(9.00)



$$x = 18$$

$$y = 0$$

$$z = 2$$

$$\frac{x+y}{z} = 9.00$$

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (C)**

$$\text{Sol. } g(x) = \begin{cases} x^2 \sin \frac{1}{x} & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$$

$$g(0) = g(0^+) = g(0^-) = 0$$

Thus $g(x)$ is continuous

$$\text{Now } g'(0^+) = \lim_{h \rightarrow 0} \frac{h^2 \sin \frac{1}{h}}{h} = 0$$

$$g'(0^-) = \lim_{h \rightarrow 0} \frac{h^2 \sin \left(-\frac{1}{h}\right)}{-h} = 0$$

$$g'(0^+) = g'(0^-) \quad [g(x) \text{ is differentiable}]$$

$$g'(x) = x^2 \cos \frac{1}{x} \times \left(-\frac{1}{x^2}\right) + \sin \left(\frac{1}{x}\right) \cdot 2x$$

$$g'(x) = -\cos \frac{1}{x} + 2x \sin \left(\frac{1}{x}\right)$$

$g'(x)$ is not continuous.

2. **Ans. (D)**

$$\text{Sol. (A)} \quad X = A + A^T$$

$$\begin{aligned} X^T &= (A^T)^T + A^T \\ &= A + A^T = X \end{aligned}$$

Nothing sure can be said about AA^T and $A^T A$

$$\text{(D) Let } X = A - A^T$$

$$\begin{aligned} X^T &= (A - A^T)^T \\ &= A^T - A \end{aligned}$$

$$X^T = -X$$

3. **Ans. (B)**

Sol. By definition of inverse

$$A^{-1} = \frac{\text{adj } A}{|A|}$$

$$(AB)^{-1} = B^{-1} A^{-1} \text{ (Reversal Law)}$$

$$(A + B)^{-1} \neq A^{-1} + B^{-1}$$

4. **Ans. (D)**

$$\text{Sol. } D_1 = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}; D_2 = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & \gamma \end{bmatrix}$$

Now we can check and say that $D_1 D_2 = D_2 D_1$

$$D_1^2 + D_2^2 = \text{Diagonal matrix}$$

$$D_1 D_2 = \text{Diagonal matrix.}$$

5. **Ans. (A)**

$$\text{Sol. } \vec{a} \times \vec{c} = \vec{b} + \lambda \vec{a}$$

$$(\vec{a} \times \vec{c}) \cdot \vec{a} = \vec{b} \cdot \vec{a} + \lambda \vec{a} \cdot \vec{a}$$

$$0 = (3 + 2) + \lambda(10)$$

$$\therefore \lambda = -\frac{5}{10} = -\frac{1}{2}$$

6. **Ans. (A)**

$$\text{Sol. } \frac{\cos A}{2R \sin A} = \frac{\cos B}{2R \sin B} = \frac{\cos C}{2R \sin C}$$

$$\cot A = \cot B = \cot C \quad \therefore A = B = C$$

Hence triangle is equilateral

$$\text{Hence area} = \frac{\sqrt{3}}{4} (a)^2 = \sqrt{3}$$

7. **Ans. (C)**

$$\text{Sol. } \tan \left(\frac{B-C}{2} \right) = \frac{b-c}{b+c} \cot \frac{A}{2}$$

$$\tan \left(\frac{B-C}{2} \right) = \frac{\sqrt{3}-1}{\sqrt{3}+1} \cot 15^\circ$$

$$\frac{B-C}{2} = 45^\circ$$

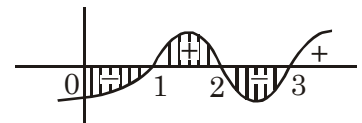
$$B-C = 90^\circ$$

$$\therefore B+C = 150^\circ$$

$$B = 120^\circ \quad C = 30^\circ$$

8. **Ans. (B)**

$$\begin{aligned} \text{Sol. Area} &= \left| \int_0^1 (x^3 - 6x^2 + 11x - 6) dx \right| \\ &+ \int_1^2 (x^3 - 6x^2 + 11x - 6) dx + \left| \int_2^3 (x^3 - 6x^2 + 11x - 6) dx \right| \\ &= \frac{11}{4} \end{aligned}$$



9. **Ans. (C)**

$$\text{Sol. } \frac{dy}{dx} = e^{9x+6} \cdot e^{-6y}$$

$$e^{6y} dy = e^{9x+6} dx$$

$$\frac{e^{6y}}{6} = \frac{e^{9x+6}}{9} + c$$

$$x = 0, y = 1$$

$$\therefore c = \frac{e^6}{18}$$

$$\text{Hence } \frac{e^{6y}}{6} = \frac{e^{9x+6}}{9} + \frac{e^6}{18}$$

$$\therefore 3e^{6y} = 2e^{9x+6} + e^6$$

10. Ans. (B)

Sol. $f(1^+) = f(1^-)$

$$\Rightarrow a + \sin^{-1}(1+b) = 1 \dots (1)$$

$$\text{Also } f'(1^+) = f'(1) = \frac{1}{\sqrt{1-(1+b)^2}} = 1$$

$$\therefore b = -1$$

$$\text{Hence : } a = 1 ; b = -1$$

11. Ans. (B)

Sol. $I = \int_a^b \frac{x^n dx}{x^n + (16-x)^n}$

$$a+b=16$$

$$I = \int_a^b \frac{(16-x)^n}{(16-x)^n + x^n} dx$$

$$2I = b-a \text{ or } b-a=12$$

12. Ans. (A)

Sol. $27\Delta^2 = s^4$

$$27s(s-a)(s-b)(s-c) = s^4$$

$$3[(s-a)(s-b)(s-c)]^{1/3} = s$$

$$\frac{s-a+s-b+s-c}{3} \geq [(s-a)(s-b)(s-c)]^{1/3}$$

$$\therefore s-a=s-b=s-c$$

$$\therefore a=b=c$$

$$\therefore a = 2R \sin A = 2 \times 4 \times \frac{\sqrt{3}}{2} = 4\sqrt{3}$$

$$\text{Area} = \frac{\sqrt{3}}{4} (4\sqrt{3})^2 = 12\sqrt{3}$$

13. Ans. (B)

Sol. $a+b+c=7$ and $|A| = abc+b$

$$\text{Let } a=2, b=3, c=2$$

$$|A|_{\max} \Rightarrow 15$$

14. Ans. (B)

Sol. $A^3 - B^3 - 3B^2C = (B+C)^3 - B^3 - 3B^2C$
 $= B^3 + 3B^2C + 3BC^2 + C^3 - B^3 - 3B^2C$
 $= 3BC^2 + C^3$
 $= 0 \quad [C^2 = 0]$

15. Ans. (A)

Sol. $\frac{dy}{dx} = \frac{(x+y)^2 + (x+y)}{(x+y)^2 + 2(x+y)}$

$$x+y=t$$

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} - 1 = \frac{t^2 + t}{t^2 + 2t}$$

$$\frac{dt}{dx} = \frac{t^2 + t}{t^2 + 2t} + 1$$

$$\frac{dt}{dx} = \frac{2t^2 + 3t}{t^2 + 2t} = \frac{2t+3}{t+2}$$

$$\frac{t+2}{2t+3} dt = dx$$

$$\frac{1}{2}t + \frac{1}{4} \ln(2t+3) = x + C$$

$$\frac{1}{2}(x+y) + \frac{1}{4}(\ln(2x+2y+3)) = x + C$$

$$\frac{y-x}{2} + \frac{1}{4} \ln(2x+2y+3) = C$$

16. Ans. (B)

Sol. $\vec{r} = \vec{b} + \lambda \vec{c}$

$$= (\hat{i} + 2\hat{j} - \hat{k}) + \lambda(\hat{i} + \hat{j} - 2\hat{k})$$

$$\vec{r} = (1+\lambda)\hat{i} + (2+\lambda)\hat{j} - (1+2\lambda)\hat{k}$$

$$\text{Now projection of } \vec{r} \text{ on } \vec{a} \text{ is } \frac{\sqrt{2}}{3}$$

$$\therefore ((1+\lambda)\hat{i} + (2+\lambda)\hat{j} - (1+2\lambda)\hat{k}) \cdot \left(\frac{2\hat{i} - \hat{j} + \hat{k}}{\sqrt{6}} \right) = \frac{\sqrt{2}}{3}$$

$$\frac{2(1+\lambda) - (2+\lambda) - (1+2\lambda)}{\sqrt{6}} = \frac{\sqrt{2}}{3}$$

$$-\lambda - 1 = 2$$

$$\boxed{\lambda = -3}$$

$$\text{Hence } \vec{r} = -2\hat{i} - \hat{j} + 5\hat{k}$$

17. Ans. (B)

Sol. $\vec{a} = \lambda \vec{b}$

$$(x_1 - 2, 2) \equiv \lambda(1, y, z)$$

$$\frac{x}{1} = -\frac{2}{y} = \frac{2}{z} = \lambda$$

$$x = \lambda, y = -\frac{2}{\lambda}; z = \frac{2}{\lambda}$$

$$\hat{n} = \frac{\vec{c}}{|\vec{c}|} = \frac{\lambda\hat{i} - \frac{2}{\lambda}\hat{j} + \frac{2}{\lambda}\hat{k}}{\sqrt{\lambda^2 + \frac{4}{\lambda^2} + \frac{4}{\lambda^2}}}$$

$$\hat{n} = \frac{1}{\sqrt{\lambda^4 + 8}}(\lambda^2, -2, 2)$$

$$\hat{n} = \frac{1}{2\sqrt{2}}(0\hat{i} - 2\hat{j} + 2\hat{k})$$

$$\lambda = 2$$

$$\hat{n} = \frac{1}{\sqrt{24}}(4\hat{i} - 2\hat{j} + 2\hat{k}) = \frac{2\hat{i} - \hat{j} + \hat{k}}{\sqrt{6}}$$

and remaining options can not be obtained for any value of λ

18. Ans. (C)

Sol. Let $P(t_1^2, t_1^3)$ and $Q(t_2^2, t_2^3)$

$$y^2 = x^3 \quad \therefore 2y \frac{dy}{dx} = 3x^2$$

$$\frac{dy}{dx} = \frac{3x^2}{2y} = \frac{3t_1^4}{2t_1^3} = \frac{3}{2}t_1$$

$$\text{Slope of PQ} = \frac{t_2^3 - t_1^3}{t_2^2 - t_1^2} = \frac{3}{2}t_1$$

$$\text{On solving } t_2 = -\frac{t_1}{2} = \frac{t_1}{t_2} = -2$$

$$\frac{\text{Slope of OP}}{\text{Slope of OQ}} = \frac{t_1}{t_2} = -2$$

19. Ans. (A)

$$\text{Sol. } f(x) = \int_{-1}^x \frac{e^{\frac{1}{x-2-t}}}{(x-2-t)^2} dt$$

$$\frac{2}{x-2-t} = y$$

$$\frac{2}{(x-2-t)^2} dt = dy \quad \text{when } t = -1, y = \frac{2}{x-1}$$

$$t = x, y = -1$$

$$\int_{\frac{2}{x-1}}^{-1} e^y dy = \frac{e}{2} (e^{-1} - e^{\frac{2}{x-1}}) = \frac{1}{2} (1 - e^{\frac{x+1}{x-1}})$$

$$\therefore f(2) = \frac{1}{2} (1 - e^3)$$

20. Ans. (A)

$$\text{Sol. Let } f(x) = \int_{\frac{1}{a}}^x \frac{\tan^{-1} t}{t e^x} dt \quad \dots (1)$$

$$\text{put } t = \frac{1}{y}$$

$$dt = -\frac{1}{y^2} dy$$

$$f(x) = \int_a^{\frac{1}{a}} \frac{\tan^{-1} \left(\frac{1}{y} \right)}{\frac{1}{y} e^x} \times \left(-\frac{1}{y^2} \right) dy$$

$$f(x) = \int_{\frac{1}{a}}^a \frac{\tan^{-1} \left(\frac{1}{y} \right)}{y e^x} dy$$

$$= \int_{\frac{1}{a}}^a \frac{\tan^{-1} \left(\frac{1}{t} \right)}{t e^x} dt \quad \dots (2)$$

$$2f(x) = \int_{\frac{1}{a}}^a \frac{2}{t e^x} dt$$

$$= \frac{\pi}{2e^x} \left[\ln a - \ln \left(\frac{1}{a} \right) \right]$$

$$2f(x) = \frac{\pi}{2e^x} \cdot 2 \ln a = \frac{\pi \ln a}{e^x}$$

$$f(x) = \frac{\pi \ln a}{2e^x}$$

SECTION-II

1. Ans. 0.50

$$\text{Sol. } e^{x+y} = y^2$$

$$x + y = 2 \ln y$$

$$1 + \frac{dy}{dx} = \frac{2}{y} \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{y}{2-y}$$

$$\frac{d^2y}{dx^2} = \frac{(2-y) \frac{dy}{dx} - y \left(-\frac{dy}{dx} \right)}{(2-y)^2}$$

$$= \frac{2}{(2-y)^2} \cdot \frac{dy}{dx} = \frac{2y}{(2-y)^3}$$

$$\therefore k = 2$$

2. Ans. 3.50

$$\text{Sol. } y = x^2 + x + 1 \quad (x = 1, y = 3)$$

$$\frac{dy}{dx} = 2x + 1$$

$$\frac{dy}{dx} = 3 \text{ at } (1, 3)$$

Equation of tangent

$$y - 3 = 3(x - 1)$$

$$y = 3x$$

Area bounded is

$$\int_{-1}^0 x^2 + x + 1 + \int_0^1 [(x^2 + x + 1) - 3x] = \frac{7}{6}$$

3. Ans. 0.50

Sol. $2b = a + c$

$$2(2R\sin B) = 2R(\sin A + \sin C)$$

$$2\sin B = 2\sin\left(\frac{A+C}{2}\right)\cos\left(\frac{A-C}{2}\right)$$

$$2\left(2\sin\frac{B}{2}\cos\frac{B}{2}\right) = 2\cos\frac{B}{2}\cdot\cos\left(\frac{A-C}{2}\right)$$

$$4\sin\frac{B}{2} = 2\cos\left(\frac{A-C}{2}\right)$$

$$\frac{2}{1} = \frac{\cos\left(\frac{A-C}{2}\right)}{\cos\left(\frac{A+C}{2}\right)}$$

$$\frac{2+1}{2-1} = \frac{2\cos\frac{A}{2}\cos\frac{C}{2}}{2\sin\frac{A}{2}\sin\frac{C}{2}}$$

$$3\tan\frac{A}{2}\tan\frac{C}{2} = 1$$

$$\frac{3}{2}\tan\frac{A}{2}\tan\frac{C}{2} = \frac{1}{2} = 0.50$$

4. Ans. 2.00

Sol. $A^{-1}B = B^{-1} \dots (1)$
 $(A^{-1}B)^{-1} = (B^{-1})^{-1}$ [Taking Inverse]

$$B^{-1}A = B$$

$$BB^{-1}A = B^2$$

$$A = B^2 \Rightarrow \therefore A = \begin{bmatrix} -2 & -2 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} -2 & -2 \\ -1 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 6 & 4 \\ 2 & 2 \end{bmatrix}$$

$$\text{Now } KA^{-1} = 2B^{-1} + I$$

$$KA^{-1}B = 2I + IB$$

$$KA^{-1}B = 2I + B \quad [\text{From (1)}]$$

$$KB^{-1} = 2I + B$$

$$KI = 2B + B^2$$

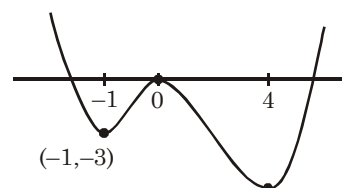
$$K = 2 \begin{bmatrix} -2 & -2 \\ -1 & 0 \end{bmatrix} + \begin{bmatrix} 6 & 4 \\ 2 & 2 \end{bmatrix}$$

$$K = 2$$

5. Ans. 0.80

Sol. $f(x) = x^4 - 4x^3 - 8x^2 = -P$

$$f'(x) = 4x(x-4)(x+1)$$



So if the equation $f(x) = -P$ has all roots real than $-3 \leq -P \leq 0$ or $0 \leq P \leq 3$

Number of integral values = 4